

Read Small-Indicator Semantics

A glow, blink, or buzz is evidence only when its meaning is bound

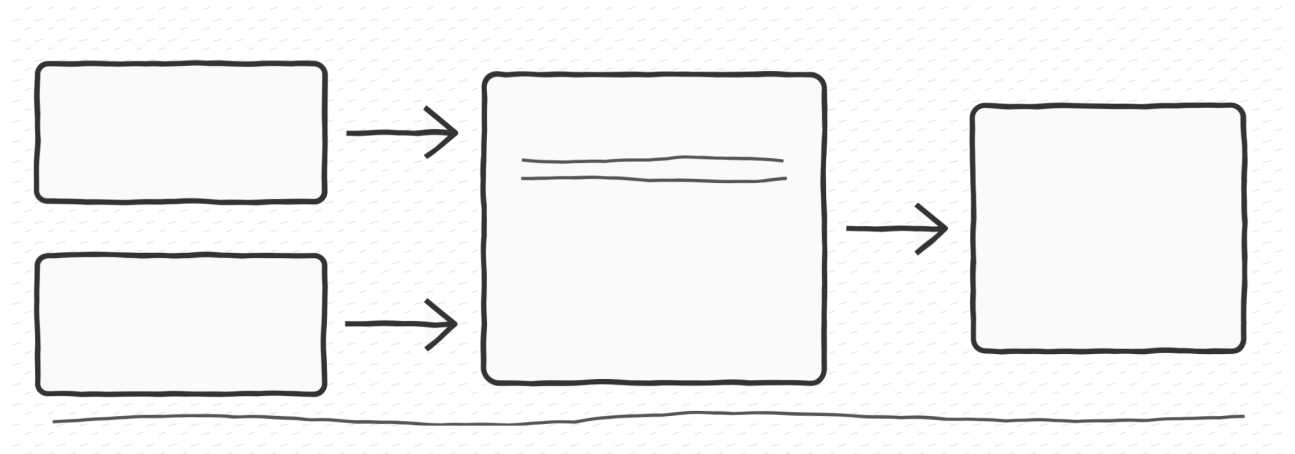
E0 COPIED-EVIDENCE REPLAY — NO PIN, LIGHT, SOUND, OR POWER CLAIM

Lesson	080 — component	Time	150–210 minutes
Prerequisite	Lesson 079 intent	Board	compile-only Mega replay
Input	copied drive, request, observation	Observation	named semantic cells

Responsibility. `SmallIndicatorSemanticsPolicy` interprets one closed indicator declaration, one copied Lesson 079 intent, one L080-owned semantic request, and one independently attributable observation. It owns no pin, endpoint, supply, waveform, scheduler, sensor, display, or storage.

1 Separate appearance from meaning

A small board that lights or sounds can still conceal its polarity, populated resistor, driver, autonomous behavior, or required safe state. A retail label does not settle those facts. E0 accepts only copied, correlated declarations and observations; it never identifies or energizes a specimen.



Alternative text: rough graphite cards represent copied drive intent and the independent request/observation evidence entering a semantic policy; a result card exits while an unpowered module remains outside E0.

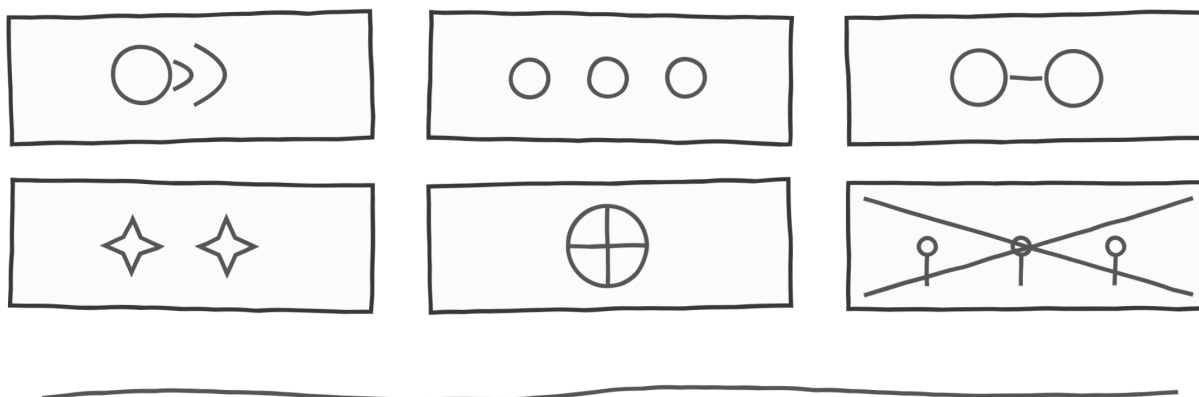
E0 establishes	E0 leaves open
closed kind and channel semantics	exact specimen identity and pin trace
copied polarity and autonomy meaning	voltage, current, brightness, or sound
freshness and correlation rules	waveform timing and physical safe state
deterministic semantic result	GPIO ownership and powered acceptance

2 Use a closed validity table

Only the six rows below are valid. The Booleans are exact declarations, not minimum requirements. Bits 6–7 are reserved zero.

Kind	Autonomy	Safe state	AH	Mask	R	D
Active Buzzer	follows	drive inactive	1	Sound	0	1
Active Buzzer	autonomous	unpowered	1	Sound	0	1
Traffic Light	follows	drive inactive	1	R/A/G	1	0
Dual Color LED	follows	drive inactive	1	R/G	1	0
Auto Flash LED	autonomous	unpowered	1	R/G/B	1	0
Voltage Indicator	observation only	high impedance	1	Voltage	1	0

AH means active high; R and D mean declared populated resistor and driver. The masks are Red = 01, Amber = 02, Green = 04, Blue = 08, Sound = 10, and Voltage = 20, all hexadecimal.



Alternative text: six hand-drawn index cards show the only accepted combinations of indicator kind, channel marks, autonomy, and safe state; a crossed-out generic three-pin card is outside the set.

Prediction: may a descriptor add Blue to a Traffic Light because the hardware happens to have another LED? _____.
Explain: _____.

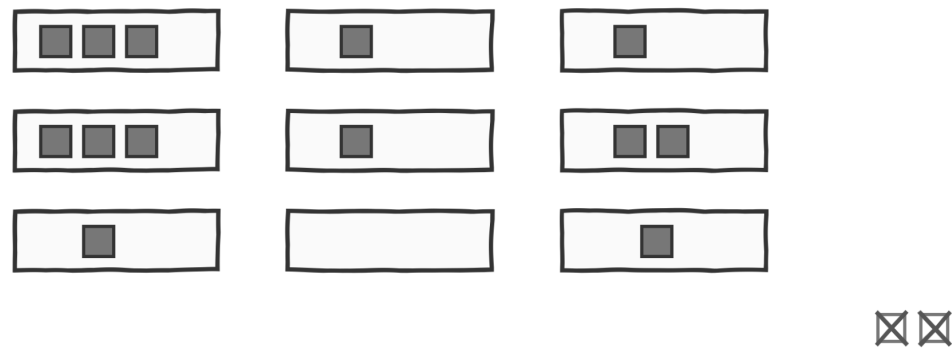
3 Bind the driver, not just its familiar fields

The copied drive must match specimen, revision, electrical evidence, policy configuration, lifecycle, session, run, step, and request correlation. It must also carry the expected full Lesson 079 descriptor digest. That digest binds family, source packet, topology, protection, and every budget field that partial comparison would miss. Digest zero is valid; it does not mean absent.

4 Select meaning before observing it

`SmallIndicatorSemanticRequest` is the L080-owned copied authority for the expected semantic selection. Its nonzero request identity is repeated by the observation and result. The selection never comes from the observation being judged: otherwise a multichannel observation could declare the color it was meant to prove.

Selected, observed, and semantic masks must be subsets of the declared mask. A follows-drive request selects exactly one declared channel while drive is active and zero while off. Autonomous requests select the complete declared mask as enable authority; their observations may report a nonzero declared subset. Observation-only requests select zero and never gain drive authority. Inactive evidence uses zero.



Alternative text: a pencil-drawn mask worksheet compares declared, request-selected, and independently observed channel squares for follows-drive, autonomous, and observation-only cases; reserved squares are crossed out.

Observation	Canonical observation fields	Semantic check
NotObserved	mask 0; level/transition/safe false; OK	incomplete
Inactive	observed 0	must agree with the semantic case
Active	nonzero declared subset	may expose semantic disagreement
Alternating	nonzero declared subset	autonomy rules classify it
Fault	mask 0; level/transition/safe false; non-OK	producer fault

Every unlisted structural combination is malformed and cannot mutate state. Within the listed Active and Alternating rows, level, transition, and safe-state values deliberately remain broad enough to expose reachable semantic disagreements. `copiedLevelHigh` should equal the nonzero mask for single-channel Buzzer and Voltage rows; a contrary value is a `PolarityMismatch`. It should be false for multichannel kinds; a true value is likewise a reachable mismatch, not erased during parsing. No mask implies a pin count or wiring topology.

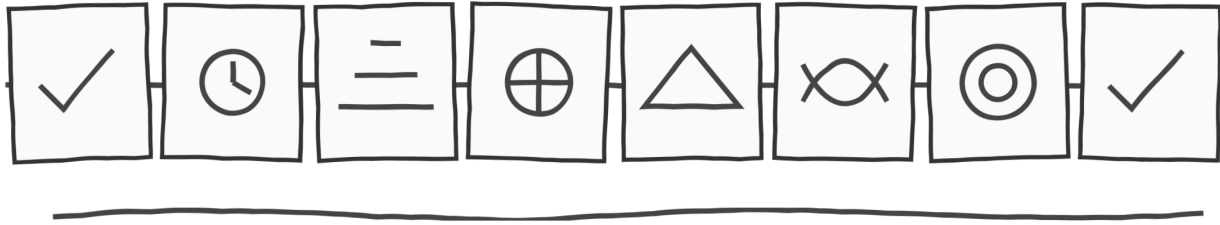
5 Keep autonomous behavior explicit

An Auto Flash LED does not become “active” merely because drive intent was admitted. Its copied observation must show a declared nonzero subset and an autonomous transition. Conversely, transition evidence on a follows-drive Traffic Light is unexpected autonomy. E0 records only that a transition was copied; cadence, duty, brightness, color accuracy, and waveform remain open.

6 Treat three streams and timing checks independently

The request and observation have distinct attributable producer streams: each has its own latched source/configuration and contiguous sequence. A shared policy sequence orders apply and cancel. The first accepted apply uses one in all three streams; each then advances independently. Wrap from maximum to zero is exhaustion, not a restart. A cancel advances only the policy stream and must use request-producer authority.

The caller supplies session start, request time, observation time, and current time. The policy reads no clock. It enforces $startedAt \leq requestedAt \leq observedAt \leq now$ with forward modular half-range comparisons. Warm-up is computed from current time minus session start; settling from observation time minus request time. No copied “satisfied” Boolean substitutes. Age equal to the declared maximum is current; one tick older is stale.



Alternative text: a graphite row of separate gate cards checks drive admission, warm-up, settling, freshness, polarity, autonomy, semantic agreement, and safe state; no card is allowed to stand in for another.

1. A safe-state observation does not prove drive admission.
2. A fresh observation does not prove warm-up or settling.
3. An admitted drive does not prove optical or acoustic behavior.
4. Missing observation is incomplete; it never means inactive.

Known-zero warm-up or settling requires no delay. Otherwise both computed checks must pass independently. Safe-state evidence is independent again: **DriveInactive**, **HighImpedanceRequired**, and **UnpoweredRequired** are semantic declarations whose physical truth still requires a later exact fixture and observed bench record.

7 Map every Lesson 079 terminal exactly

Only these Lesson 079 state/reason pairs are supported; activity, output level, and producer status must also be canonical. Every unlisted tuple is malformed rather than silently generalized.

Drive state/reason	L080 result before observation semantics
Off/None	evaluate only a zero selection
Off/Expired	Rejected/DriverExpired
Requested/None	evaluate nonzero follows/autonomous selection
Rejected/SourceIneligible	Rejected/SourceIneligible
Rejected/BudgetExceeded	Rejected/DriverBudgetExceeded
Rejected/BaseBudgetInsufficient	Rejected/DriverBaseBudgetInsufficient
Rejected/FlybackMissing	Rejected/DriverFlybackMissing
Rejected/SequenceDiscontinuity	Rejected/DriverSequenceDiscontinuity
Rejected/TimestampDiscontinuity	Rejected/DriverTimestampDiscontinuity
Rejected/CapacityExceeded	Rejected/DriverCapacityExceeded
Cancelled/Cancelled	Cancelled/DriverCancelled
Fault/ProducerFault	ProducerFault/ProducerFault
Fault/Cancelled	ProducerFault/DriverCancelled
Shutdown/None	Shutdown/DriverShutdown

8 Apply failure precedence

Order	Condition	Result
1	malformed structure or enum	API rejection; no mutation
2	lifecycle or identity mismatch	API rejection; no mutation
3	cancel or shutdown	all-off/cancel path
4	request, drive, observation producer failure	first attributable fault
5	mapped drive terminal	exact named result above
6	stream/time discontinuity or stale	named rejection
7	warm-up, settling, missing observation	incomplete result
8	polarity/autonomy/mask disagreement	named semantic rejection
9	safe-state disagreement	SafeStateMismatch
10	complete matching evidence	accepted result

9 Replay without energizing a module

The Mega sketch builds copied descriptors and fixtures for all five concrete kinds. It applies correlated drive intents, semantic requests, and observations, then copies results into named volatile cells. It performs no GPIO, timing, waveform, optical, acoustic, or supply operation.

1. **Predict** disposition, reason, semantic mask, autonomy flag, and safe-state flag before each call.
2. **Observe** named volatile cells with a debugger, simulator, or disassembly. Serial is optional and never sole evidence.
3. **Interpret** agreement as deterministic E0 semantic evidence, never as proof that a physical indicator lit, blinked, or sounded.

Atomicity experiment. Place recognizable canaries around the output result. Submit a changed duplicate, digest mismatch, reserved mask bit, a gap in each of the three streams, future time, and malformed observation. Confirm that each structural or correlation API rejection preserves the policy, output, and canaries. Contrast a terminal chronology or well-formed semantic disagreement, which commits an attributable rejected result for byte-identical replay.

10 Diagnose copied evidence

Symptom	Inspect first	Interpretation
non-OK API status	encoding, lifecycle, full correlation	no result mutation
DriveNotAdmitted	Lesson 079 state/reason	semantics gain no authority
WarmupUnsatisfied	start and supplied observation time	evidence arrived too soon
Stale	observation age boundary	copied evidence is too old
AutonomousWaveformMissing	transition flag and state	autonomous claim lacks change evidence
UnexpectedAutonomy	descriptor row and transition	follows-drive behavior disagrees
SafeStateMismatch	distinct safe-state observation	another check cannot substitute
ObservationMismatch	selected/observed masks	channel meaning disagrees

11 Test the complete contract

- Exhaust all enum encodings and every kind/autonomy/safe-state/polarity/ population/mask cross-product.
- Exercise exact request selections for all five fixtures, every structurally attributable observation row, and every reachable level/transition/safe-state disagreement.
- Mutate every driver correlation field and the full descriptor digest; retain literal digest goldens independent of production code.
- Test computed warm-up/settling, freshness equality/one-over, rollover, half-range time, all three streams, independently latched request and observation sources, duplicates, gaps, cancellation, and shutdown.
- Test every canonical and malformed Lesson 079 mapping tuple.
- Cross producer, stop, stale, warm-up, settling, polarity, autonomy, observation, and safe-state failures at every precedence boundary.

12 Open acceptance cards

E0 software	Pending final host/sanitizer, style/header, Mega, exact resource, independent review, publication, and terminal architecture gates.
E1 identity	Open: both-face photographs, markings, primary sources, traced pin order/topology, passive measurements, energy class, and reviewed exact descriptor. E1 includes no powered observation.
E2a/E2b fixture	Open: exact indicator/driver/adaptor, authoritative schematic, limits, resource map, non-Serial observation, rollback, independent power removal, safe-state proof, and recorded bench acceptance.
Physical result	Open. This PDF contains no circuit, powered observation, supported-specimen claim, or physical safe-state result.

13 Resource boundary pending final measurement

The controlling exact targets are 14 KiB flash, 1,024 B static SRAM, 448 B synchronous stack, a 384 B policy, and a 256 B evidence value; hard limits are 20 KiB, 1,536 B, 640 B, 512 B, and 384 B respectively. Final ordinary and exact values, fingerprint, and any independent target-miss disposition must replace this provisional paragraph before Lesson 080 promotion. No estimated number is presented as measurement.

14 Exercises

1. Explain why a passing Traffic Light mask says nothing about pin count.
2. Construct one autonomous observation that is structurally valid but yields `AutonomousWaveformMissing`.
3. Explain why the full Lesson 079 digest is required in addition to inspectable specimen and policy fields.
4. Explain why request and observation sequences cannot be collapsed into the policy operation sequence.

Acceptance record — leave open until observed.

Exact specimen and revision: _____

Primary-source packet and reviewer: _____

Fixture, current limit, and independent power removal: _____
--

Resource-acquisition observation: _____

Independent visual/acoustic path and safe-state observation: _____
--

Bench date, result, and reviewer: _____

15 Promotion boundary

Before publication, replace provisional resources with exact fingerprint-bound evidence, complete independent terminal review, re-read the work queue, and confirm that Lesson 081 consumes this policy without hidden ownership, identity exceptions, duplicated lifecycle rules, or relaxed semantics. Any such strain requires a bounded design remediation, not another option flag.